

MLFF SIZE-HALVE1 Work-Exposure and Optimization-Coverage Analysis

CPU-testable planning evidence for the corrected 3/10/30 target-size funnel

mdstats project

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1 Scope

This report evaluates the corrected target-size funnel without claiming MACE/GPU training performance. Coverage is hard admission only; all qualified sizes enter the 3-epoch screen. The exposure model is exact for structure-epoch counts under the frozen 3/10/30 policy, but it is not a wall-time model.

2 Frozen funnel

$$N_{\text{eligible}} \xrightarrow{3 \text{ epochs}} \leq 4 \xrightarrow{10 \text{ epochs}} 2 \xrightarrow{30 \text{ epochs}} 1.$$

Policy digest: 49bf6d52e0ce78eb06176f14d534afb6d016179e9b90d7696a752f8bb49cb4f8.

The target structure-epoch exposure proxy is

$$W = 3 \sum_{i \in A} K_i + 7 \sum_{i \in S_4} K_i + 20 \sum_{i \in S_2} K_i,$$

relative to

$$W_{\text{full}} = 30 \sum_{i \in A} K_i.$$

3 Exposure envelope

The table enumerates monotone suffixes of the default nested ladder, from the minimum three qualifiers through all seven. **Best** means the smallest permitted survivors continue; **worst** means the largest permitted survivors continue.

Qualifiers	Eligible sizes	Best reduction	Worst reduction
3	2048, 4096, 8192	38.10%	9.52%
4	1024, 2048, 4096, 8192	53.33%	13.33%
5	512, 1024, 2048, 4096, 8192	72.26%	15.81%
6	256, 512, 1024, 2048, 4096, 8192	81.27%	16.98%
7	128, 256, 512, 1024, 2048, 4096, 8192	85.67%	17.56%

With all seven default sizes admitted, the exposure reduction is bounded between **17.56%** and **85.67%**, depending on which sizes survive. This wide range is why PERF-P2R must benchmark both unfavorable large-size-survivor and favorable small-size-survivor cases rather than report one convenient path.

4 Required qualification additions

1. **SIZE-FIDELITY1 before performance promotion.** Exhaustively continue all hard-coverage qualifiers to 30 epochs for multiple screening seeds; retrospectively test epoch-3 top-four and epoch-10 top-two recall of the eventual 30-epoch winner/finalists.
2. **Coarse-monitor calibration.** Compare the fixed coarse role against the full development role at epoch 3 and choose the smallest monitor that preserves promotion decisions up to practical equivalence. The current 256-frame setting is provisional.
3. **Early-equivalence calibration.** Calibrate the coarse practical-equivalence width from real trajectory/seed variability. A tied largest boundary is preserved within its band so bounded-ladder nonconvergence remains observable.
4. **Sampler-aware continuation.** Exact 3->10->30 continuation must include DataLoader/sampler/worker ordering state or prove deterministic epoch-boundary reconstruction, in addition to model, optimizer/scheduler, and global RNG state.
5. **Worst-case performance matrix.** PERF-P2R must cover 3, 4, 5, 6, and 7 admitted sizes and both small-survivor and large-survivor extremes, plus single- and multi-GPU resource regimes where available.
6. **Fuse execution, not authority.** Boundary evaluation may run in the same process as training to avoid checkpoint reload only when it emits the same separate EVAL2 scientific evidence. Shared graph/preprocessing caches and nested prefix manifests must remain byte/array equivalent.
7. **Checkpoint-I/O control.** Full restart authority is mandatory at epochs 3, 10, and 30. Additional recovery checkpoints may use a bounded execution-only cadence and may be reclaimed after immutable elimination evidence is frozen.

5 Environment and limitation

This analysis ran under **AMD EPYC 9V74 80-Core Processor** with cgroup CPU limit 800000 100000 and memory limit 4294967296. No authorizing MACE runtime/checkpoint or GPU was available. Therefore no epoch-3/10/30 wall-time, GPU-utilization, VRAM, or survivor-fidelity claim is made here.

6 References

The staged budget-allocation pattern is related to successive-halving and Hyperband, but mdstats keeps its own deterministic scientific metrics, hard coverage gate, exact continuation, and provenance contract.

1. Kevin Jamieson and Ameet Talwalkar, *Non-stochastic Best Arm Identification and Hyperparameter Optimization*, PMLR 51, 2016. PMLR article.
2. Lisha Li, Kevin Jamieson, Giulia DeSalvo, Afshin Rostamizadeh, and Ameet Talwalkar, *Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization*, JMLR 18(185), 2018. JMLR article.